

Banmeet Kaur

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EDUCATION

Hofstra University

September 2021 - May 2025

Bachelor of Science in Computer Science

GPA: 3.84

- Relevant Coursework: Machine Learning, Artificial Intelligence, Probability & Statistics, Software Engineering, Data Structures and Algorithms, Database Management, Programming Languages, and Linear Algebra
- Member of Upsilon Pi Epsilon (UPE), the International Honor Society for the Computing and Information Disciplines

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, C++, C#, TypeScript, Dart, Perl, F#

Frameworks & Libraries: Scikit-learn, TensorFlow, Hugging Face Transformers, NumPy, Pandas, PyTorch, Matplotlib, Seaborn, Flask, React/React Native, Flutter, Django

Tools & Platforms: Git, REST APIs, PostgreSQL, Tableau, PowerBI

EXPERIENCE

Hofstra University, Hempstead, NY – Computer Science Tutor

February – May 2025

- Provided one-on-one tutoring to help students master programming, data structures & algorithms, and foundational computer science concepts, resulting in improved problem-solving skills and higher academic performance

Cognizant, New York, NY – Generative AI Externship

June – August 2024

- Increased model accuracy from 9.05% to 91.95% and reduced evaluation loss from 5.24% to 0.25% through fine-tuning GPT-2 using LoRA for parameter-efficient fine-tuning with Hugging Face Transformers
- Achieved a 100% accuracy in distinguishing dog vs. non-dog images across three different Convolutional Neural Network (CNN) architectures in a pre-trained image classifier project

Hofstra University, Hempstead, NY – Advanced Summer Program in Research

June – August 2024

- Designed an Artificial Intelligence (AI) agent using Ollama to facilitate a chat experience for students and prospective students, improving access to University information
- Methodically examined the impact of document formats and layouts on retrieval-augmented generation (RAG) systems, proposing optimal formats for various combinations of embedding models and language models
- Agent is currently in beta testing for wider release to the University
- Presented all findings and work to Engineering Dean and Chairs of all Engineering departments

PROJECTS

Machine Learning - JobMate

- Led the development of JobMate, an AI-powered browser extension that streamlines job applications by auto-filling fields
- Designed ML-driven system to compare job descriptions with user resume for compatibility scoring, utilizing Hugging Face Transformers to generate a matching score
- Developed a phishing detection algorithm using Scikit-learn to protect users from suspicious job postings
- Enabled efficient extraction of key information using Natural Language Processing (NLP) libraries for resume parsing

Data Science - Formula 1 Result Predictor

- Conducted in-depth analysis into historical and current Formula 1 data, including circuit details, weather conditions, team statistics, and driver performance
- Utilized Exploratory Data Analysis (EDA) and data cleaning/preprocessing techniques, including StandardScaler and OneHotEncoder, to ensure optimal model performance
- Achieved an accuracy score of 84% with a custom machine learning model developed using Python, Pandas, and Scikit-learn
- Designed and published an interactive Tableau dashboard visualizing key Formula 1 race statistics to communicate insights on drivers, teams, and tracks

Software Engineering - FitHub

- Led the development of FitHub, a comprehensive iOS application designed to assist users in tracking and managing their fitness journey, utilizing TypeScript/React Native for the frontend and Python/Django for the backend
- Implemented functionalities including a user account system, friend connections, data visualization/analysis, and PostgreSQL integration for secure data storage

Mobile App Development - ReadCraft

- Created a book browsing application using Flutter/Dart for the frontend and Python/Flask with PostgreSQL for the backend, providing features such as user authentication and chat functionality
- Implemented a custom recommendation engine using collaborative filtering with k-Nearest Neighbors (KNN) to offer personalized book suggestions based on similar user preferences, boosting suggestions for users with similar ages